



A CFD coupled photo-bioreactive transport modelling of tubular photobioreactor mixed by peristaltic pump

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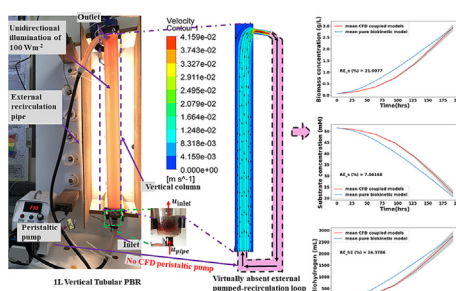
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HIGHLIGHTS

- Externally pumped-recirculated photobioreactors (PBRs) proffers pure biohydrogen.
- The PBR's CFD modelling framework under uncertainty was presented herein.
- The framework's computational cost requirements and savings were detailed.
- Combined accelerated growth kinetics and parallel computing reduces compute times.
- Enhanced radial mixing was suggested to mitigate stagnant near-wall regions.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 10 September 2022

Received in revised form 22 January 2023

Accepted 25 January 2023

Available online 8 February 2023

Keywords:

CFD modelling
Tubular photobioreactor
Biokinetic modelling
Biohydrogen production
Uncertainty analysis

ABSTRACT

Optimisation and upscaling of biohydrogen production in externally pumped-recirculated photobioreactors (PBRs) are hampered by the infeasible computational cost of coupling hydrodynamics, to light transmission and bioreaction transport, through the column and peristaltic pump. This study approximates the complexities of peristaltic pump with inlet–outlet boundary conditions (BCs), introduces mean residence time (t_m) for updating BCs, and unifies t_m -dependent accelerated growth kinetics to parallelised Computational Fluid Dynamic (CFD) environment. Due to superior cost-savings compared to previous CFD studies, the bioprocess reliability under 5% coupled CFD-photo-bioreaction parameter uncertainties was investigated for the first time, and thoroughly validated with literature bioreaction data for a 1L PBR. The results agreed to within 10% of error for simulated velocities, identifying undesired regions with poor radial mixing, and showed similar output uncertainties between the coupled CFD-photo-bioreactions and pure photo-bioreactions models, indicating absence of numerical diffusion. Therefore, this approach has great potentials for modelling other similar biosystems.

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1. Introduction

Microbial cultivation in photobioreactors (PBRs) for high-value bio-based products, fuels, and materials presents a viable option for tackling the ever-increasing mass and energy demands of the

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world's rapid population growth rate. Hence, these biotechnologies are receiving increasing interest from academia and the biorefinery, pharmaceutical, cosmetic and aquacultural industries (Aslanbay Guler et al., 2019; Pires et al., 2017). Computational Fluid Dynamics (CFD) is a powerful numerical tool for accelerating the development and optimisation of novel biotechnologies, reducing the labour and expenses required for the physical design (Pires et al., 2017; Wang et al., 2021). Experimentally validated CFD simulations have been extensively applied in PBR design, upscaling and structural optimisation of different configurations including: closed-PBRs: (i) continuous stirred tank PBRs (Zhang, 2013; Delafosse et al., 2015; Elqotbi et al., 2013), (ii) torus PBRs (Pruvost et al., 2006), (iii) flat-plate PBRs (Wang et al., 2021; Anye Cho et al., 2021; Li et al., 2014), (iv) airlift driven tubular PBRs (Luo and Al-Dahhan, 2011; Huang et al., 2016), (v) bubble column tubular PBRs (Gao et al., 2018; Nauha and Alopaeus, 2013; Liu and Luo, 2019; López-Rosales et al., 2019), (vi) thermosiphon PBRs (Cho and Pott, 2019), (vii) annular PBRs (Basak et al., 2016), (viii) semi-partitioning PBRs (Teke et al., 2022), and (ix) taylor-couette PBRs (Papacek et al., 2018; Gao et al., 2017; Gao et al., 2017), to name a few.

Among the closed-PBR configurations, airlift and bubble column driven tubular PBRs are the commonly preferred configurations due to their (i) high surface-area-to-volume ratio, (ii) flexible tube arrangement enabling different patterns and orientation to maximise the photosynthetic efficiency, (iii) low energy consumption, and (iv) efficient mixing with low shear rate (Aslanbay Guler et al., 2019; Trentin et al., 2020). However, as an air-liquid-solid multiphase system, they are not attractive for producing gaseous biofuels such as biohydrogen due to the added expenses of separating the gaseous biofuel from the air mixture during downstream processing (Skjånes et al., 2016; Kunjapur and Eldridge, 2010). As a result, several modifications have been proposed in the literature (Skjånes et al., 2016; Ross and Pott, 2021; Palamae et al., 2018), including the addition of (i) an external recirculation loop, (ii) a biohydrogen collection unit, and (iii) a peristaltic pump to provide airless pumped-recirculation mixing for a biohydrogen gas-liquid-solid tri-phasic system, labelled herein as an externally pumped-recirculated tubular PBR.

Although the externally pumped-recirculated tubular PBR is an attractive biotechnology to be upscaled due to its higher biohydrogen productivity and percentage purity (e.g., $94.1 \pm 0.4\%$ in (Ross and Pott, 2021), CFD modelling of the configuration is limited to only a few studies in literature. For instance, the authors El Maakoul et al., (El Ibrahim et al., 2020) replaced the external recirculation loop within their CFD model with a mass flowrate inlet and outlet boundary conditions to approximate the pumped-recirculation action for a 90 L anaerobic digester used for biogas production. They then investigated the effects of increasing recirculation rate on heat delivery and pressure drop. Similarly, Saini et al., (Saini et al., 2021) numerically investigated the optimum recirculation rate for complete mixing of the anaerobic digester at a reduced power consumption cost. On the other hand, the authors Rivas et al., (Rivas et al., 2020) used a velocity inlet and pressure outlet for the recirculation boundary conditions to investigate the spatial and temporal distribution of scalar tracers (i.e., mole fraction of methylene blue, but could be the concentration of any arbitrary specie for other investigations) in a (i) water and methylene blue, and (ii) water, methylene blue and artificial biomass mixtures within a 2.02 L Plexiglas tubular PBR. To guarantee recirculation of the simulated scalar, the scalar value at the outlet section was calculated and then used to update the inlet scalar value every 10 mins for the duration of a 200 min CFD simulation.

Despite attaining satisfactory results, these studies were mainly focused on the fluid flow and mixing visualisation, while microbial growth, substrate consumption and biogas production were omit-

ted from their CFD models. Hence, more robust CFD models are needed to capture the interacting effects of both the biotic and abiotic factors within such complex PBR biosystems. Specific to the scalar inlet updating time interval of 10 mins in Rivas et al., (Rivas et al., 2020), the reasons for choosing this time changes to adequately represent the recirculation process of any arbitrary PBR were not provided. Thus, it is still unclear how to choose an updating time for an externally pumped-recirculated tubular PBRs of different scales and configurations. On the other hand, mean residence time has often been exploited in the CFD modelling of photocatalytic reactors (Vaiano et al., 2015; Zhao et al., 2016) and chemical reactors (Wols et al., 2010; Greene et al., 2006) to analyse the degree of contaminant reduction and microbial inactivation. Therefore, studies which utilises the mean residence time for informing the frequency at which the scalar inlet-outlet boundary conditions are to be updated, remains to be elucidated in the CFD modelling of replaced external recirculation loop with inlet-outlet boundary conditions.

Our previous work demonstrated that accelerating the simulated growth kinetics (Anye Cho et al., 2021) is an efficient strategy for coupling the hydrodynamics and scalar transport phenomena, reducing the CFD computational clock time from a few days to a few hours without adversely affecting simulation accuracy. Meanwhile, parallel computing also reduces the computational cost immensely; for instance, Nauha and Alopaeus (Nauha and Alopaeus, 2015) parallelised their CFD simulation over 6 processors, reducing the computational clock time needed for simulating a 60 L bubble column from a few months to a few days. Therefore, merging both approaches present a unique opportunity to address the computational burden of coupling expensive CFD simulations with multiple state variables dynamics. In particular, estimation of prediction uncertainties from the biokinetic model parameters coupled to the CFD hydrodynamic model is of utmost interest but was previously unexplored due to the prohibitively high computational cost of Monte Carlo analysis.

Therefore, this manuscript aims at (i) to develop a CFD coupled hydrodynamics, light transport and biokinetic model for an externally pumped-recirculated tubular PBR, (ii) to validate the model against a literature case study (Anye Cho et al., 2021) for biomass growth, glycerol consumption and biohydrogen production by the photosynthetic bacterium *Rhodospseudomonas palustris* NCIMB 11774, and (iii) for the first time, combine the computational cost-savings achieved by accelerated simulated growth kinetics and parallel computing to investigate the reliability of the CFD coupled hydrodynamic-biokinetic model via uncertainty propagation with Monte Carlo analysis.

2. Methodology

2.1. Experimental setup and tracer dye studies

The operational mode of the VT-PBR has been detailed in our previous works (Ross and Pott, 2021; Anye Cho et al., 2021); thus, only the adaptation of the experimental setup for tracer dye studies was presented herein. For this, ten monitoring points, labelled P_{1-10} , each 0.05 m apart were marked on the vertical column, as shown in A) of Fig. 1. Upon reaching hydrodynamic steady-state, achieved by recirculating phenolphthalein indicator in distilled water through the VT-PBR for at least one minute, 3 mL of 3.5 M NaOH was injected into the flow field via a sampling port in the external recirculation pipe producing a pink coloration. As the pink tracer was carried upward from P_1 to P_{10} , the time taken for the pink colouration to pass each monitoring point was recorded. By dividing the distance between the monitoring points by the time taken to pass each point, the local velocity of the flow field was

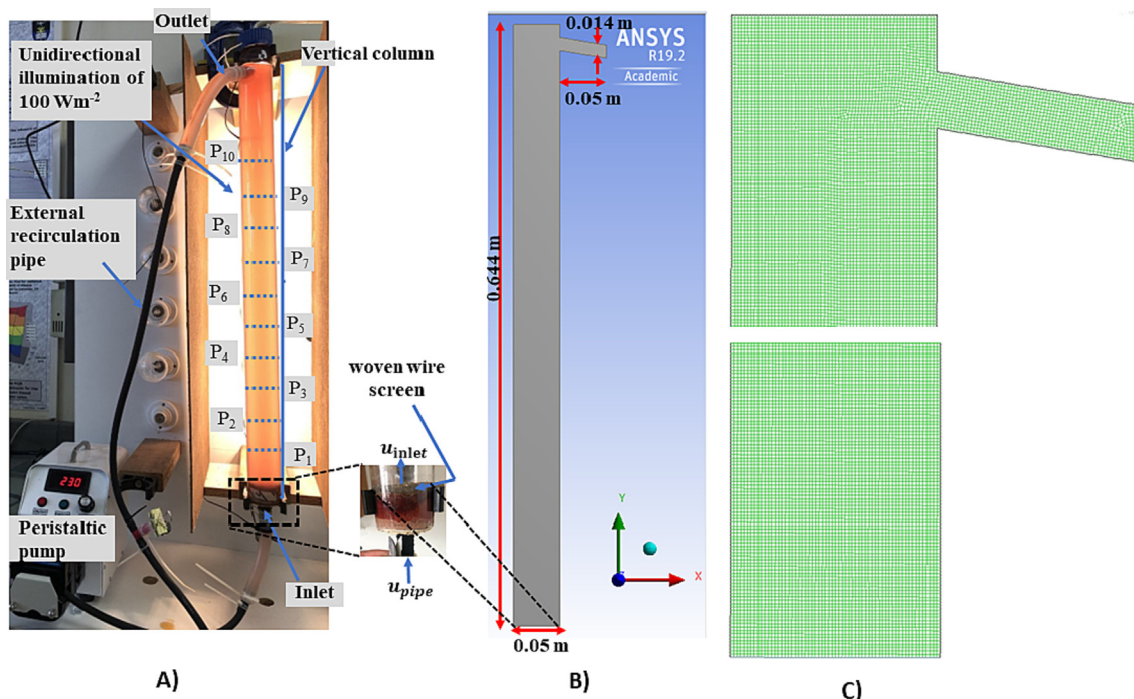


Fig. 1. Experimental and CFD step up. A) Vertical tubular photobioreactor (VT-PBR) whereby P_{1-10} are local velocity measuring points for the tracer dye study, B) 2D VT-PBR geometry with dimensions, and C) Quadrilateral dominant grid elements used in the domain meshing.

determined. This procedure was repeated in triplicate, and the data collected was later used for validation of the CFD model's prediction at the locations of the numerical monitoring points.

2.2. Computational fluid dynamics (CFD)

2.2.1. Hydrodynamic model and CFD domain

As illustrated in A) of Fig. 1, the VT-PBR was characterised by three main sections: i) a vertical column illuminated by 100 W m^{-2} bulbs, ii) a dark (i.e., non-illuminated) external recirculation pipe, and iii) a peristaltic pump for the pumped-recirculation mixing. While each section is important, a CFD model that included them all would be too computationally expensive to simulate. Hence, the VT-PBR CFD model made the following simplifying assumptions:

- Only the illuminated vertical column and outlet section were included in the CFD model, while the dark recirculation pipe and peristaltic pump were omitted as they were considered photosynthetic dead zones without light.
- The 3D geometry of the remaining illuminated column was approximated with the 2D design shown in B) of Fig. 1 due to the symmetrical tubular nature of the VT-PBR.
- Flow velocity via the inlet and outlet sections of the 2D geometry were captured with suitable boundary conditions, as reported in Table 1. Therefore, the peristaltic pump was omitted from the CFD model like in other literature sources (El Ibrahimy et al., 2020; Saini et al., 2021; Rivas et al., 2020).
- The smaller pipe's inlet velocity (i.e., u_{pipe}) was uniformly distributed by the woven wire screen into the VT-PBR's inlet velocity (i.e., u_{inlet}). However, simulating its smaller grid element sizes would significantly increase the number of elements count and thus computational cost. Therefore, mass balance was applied to estimate u_{inlet} with Eq. (1a) whereby A_{pipe} and $A_{\text{VT-PBR}}$ are the cross-sectional areas of the smaller pipe and the VT-PBR respectively.

Table 1

Operating conditions of the VT-PBR.

General boundary conditions	Hydrodynamic model	Mass (scalar) transport model
Inlet conditions	Velocity inlet (m s^{-1}) $u_{\text{inlet}} = 0.0055$	Dirichlet (updated after every mean residence time)
Outlet conditions	Pressure outlet	Neuman (i.e., zero flux)
Top wall condition (water surface)	Free surface of constant shear (free slip) at 1 atm	Neuman (i.e., zero flux)
All other wall conditions	Non-slip	Neuman (i.e., zero flux)

$$u_{\text{inlet}} = \frac{A_{\text{pipe}} \cdot u_{\text{pipe}}}{A_{\text{VT-PBR}}} \quad (1a)$$

- The Reynolds number was calculated to be 381.9, indicating a laminar flow regime within the vertical column, thus Eq. (1b) for fully developed laminar flow profile was implemented as the inlet boundary condition of the CFD's hydrodynamic model.

$$u(r) = 2 \cdot u_{\text{inlet}} \cdot \left[1 - \left(\frac{r}{R} \right)^2 \right] \quad (1b)$$

whereby $u(r)$ is the radial velocity, $2 \cdot u_{\text{inlet}}$ is the maximum velocity occurring at the centre when the radius $r = 0$, R is the inner radius of the VT-PBR.

- The absence of visible biohydrogen bubbles during the operation of the experimental setup and the assumption of density similarities between cell biomass and water (Anye Cho et al., 2021; Wols et al., 2010) reduces the investigation into a single-phase flow problem.

- Fluid within the entire CFD domain was incompressible, Newtonian, and operating under isothermal conditions at 301.15 K; thus, heat exchange was not considered.

Based on these assumptions, the continuity (i.e., Eq. (2)) and momentum (i.e., Eq. (3)) equations were solved as the CFD hydrodynamic model.

$$\frac{\partial \rho}{\partial t} = \nabla \cdot (\rho \mathbf{u}) = 0 \quad (2)$$

$$\frac{\partial}{\partial t}(\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) = -\nabla P + \nabla \cdot \left(\mu \left[(\nabla \mathbf{u} + \nabla \mathbf{u}^T) - \frac{2}{3} \nabla \cdot \mathbf{u} \mathbf{I} \right] \right) + \rho \mathbf{g} + \mathbf{F} \quad (3)$$

where $\mathbf{u}$ is the velocity, ρ is the density, while $\rho \mathbf{g}$ and $\mathbf{F}$ are the gravitational body force and external body forces, respectively, P is the static pressure, μ is the molecular viscosity, and $\mathbf{I}$ is the unit tensor.

2.2.2. Biokinetic model

Our recent work proposed a biokinetic model for simulating biomass growth, substrate consumption and biohydrogen production by the photosynthetic bacterium *R. palustris* within the VT-PBR (Anye Cho et al., 2021); reproduced in Eq. (4) to Eq. (6). However, to approximate the biokinetics of the VT-PBR based solely on the productivity of the vertical column, it was necessary to correct for the fraction of time, T_L the bacteria cells spent within the illuminated zone of the experimental VT-PBR, as defined by Eq. (7). A caveat herein is that photosynthetic biomass growth was not limited by substrate concentration as it was supplied in replete amounts, hence the maintenance term in Eq. (5) was not influenced by T_L .

$$\frac{dX}{dt} = T_L \cdot A' \cdot \exp\left(-\frac{E_a}{RT}\right) \cdot \left(\frac{\frac{I_0}{(\tau X)L} \cdot (1 - e^{-(\tau X)L})}{k_s \left(\frac{1}{\eta}\right) + \left(\frac{I_0}{(\tau X)L} \cdot (1 - e^{-(\tau X)L})\right)} \right) \cdot X \quad (4)$$

$$\frac{dS}{dt} = -Y_{XS} \cdot \left(T_L \cdot A' \cdot \exp\left(-\frac{E_a}{RT}\right) \cdot \left(\frac{\frac{I_0}{(\tau X)L} \cdot (1 - e^{-(\tau X)L})}{k_s \left(\frac{1}{\eta}\right) + \left(\frac{I_0}{(\tau X)L} \cdot (1 - e^{-(\tau X)L})\right)} \right) \cdot X \right) - m \cdot X \quad (5)$$

$$\frac{dH_2}{dt} = T_L \cdot \phi \cdot A'_{H_2} \cdot \exp\left(-\frac{E_{a,H_2}}{RT}\right) \cdot \left(\frac{\frac{I_0}{(\tau X)L} \cdot (1 - e^{-(\tau X)L})}{k_{s,H_2} \left(\frac{1}{\eta}\right) + \left(\frac{I_0}{(\tau X)L} \cdot (1 - e^{-(\tau X)L})\right)} \right) \cdot X \quad (6)$$

$$T_L = \frac{t_L}{t_L + t_D} \quad (7)$$

whereby A' ($8.30 \times 10^3 \text{ h}^{-1}$) and A'_{H_2} ($1.01 \times 10^{10} \text{ h}^{-1}$) are the pre-exponential factors for biomass growth and biohydrogen production, E_a ($2.74 \times 10^4 \text{ J mol}^{-1}$) and E_{a,H_2} ($4.78 \times 10^4 \text{ J mol}^{-1}$) are the activation energies for biomass growth and biohydrogen production, R ($8.3145 \text{ J mol}^{-1} \text{ K}^{-1}$) is the universal gas constant, T (301.15 K) is the absolute temperature, I_0 (100 W m^{-2}) is the incident light intensity, k_s (500.0 W m^{-2}) and k_{s,H_2} (500.0 W m^{-2}) are the light saturation coefficients for biomass growth and biohydrogen production, τ ($90.8 \text{ m}^2 \text{ g}^{-1}$) is the light absorption coefficient, L (0.05 m) is the light path length, Y_{XS} (9.66 mmol g^{-1}) and m ($0.0140 \text{ mmol g}^{-1}$) are the substrate yield coefficient and maintenance coefficient, η (4.515) is the effective light coefficient and ϕ (1.945), is the biohydrogen enhancement factor. Finally, t_L (s) and t_D (s) are the lengths of time spent in the light and dark zones, respectively, as estimated from the CFD model using Eq. (8) based

on the predicted flow rates $\dot{V}_i$ through the illuminated vertical column (i.e., $i = L$) of volume V_L and dark recirculation pipe (i.e., $i = D$) of volume V_D .

$$t_i = \frac{V_i}{\dot{V}_i} \quad (8)$$

2.2.3. Coupling of CFD hydrodynamics and biokinetic models

To integrate the dynamics of the biokinetics and concentration distribution within the VT-PBR, the three mass (scalar) transport equations describing the bioreaction species corresponding to the biomass growth (i.e., Eq. (9)), substrate consumption (i.e., Eq. (10)) and biohydrogen production (Eq. (11)) were coupled with the converged velocity flow fields from solving Eq. (2) and Eq. (3). On the left-hand side of Eqs. (9)–(11), the first and second terms denote the accumulation and convection of the bioreaction specie, respectively. On the right-hand side of Eqs. (9)–(11), the first and second terms denote bioreaction specie diffusion and growth, respectively, where the diffusivities were taken from literature: $D_{e,X} = (5.5 \times 10^{-10} \text{ m}^2 \text{ s}^{-1})$ (Anye Cho et al., 2021); $D_{e,S} = (1.06 \times 10^{-9} \text{ m}^2 \text{ s}^{-1})$ (Haynes et al., 2017) and $D_{e,H_2} = (5.11 \times 10^{-9} \text{ m}^2 \text{ s}^{-1})$ (Haynes et al., 2017). The CFD implementation of the light transport model (i.e., $I_0 \cdot \exp(-\tau X L)$) within the specie growth terms has already been detailed in our previous work (Anye Cho et al., 2021) but was modified herein as illustrated in Table 2. This was to eliminate interpolation errors between the scalar's currently occupied mesh cell elements and the unoccupied neighboring cell elements of the domain. The modification guarantees the strong coupling of the column's rising plume to photobioreactions by tracking the rising plume's vertical distance and activating the light transport model in a compartmentalised approach. Since no significant improvement in the simulation results were observed beyond 50 compartments (i.e., 0.0128775 m each), thus 50 was deemed satisfactory to avoid higher computational costs.

$$\frac{dX}{dt} + [\nabla \cdot (\mathbf{u}X)] = \nabla \cdot (D_{e,X} \nabla X) + \left[A' \cdot \exp\left(-\frac{E_a}{RT}\right) \cdot \frac{I_0 \cdot \exp(-\tau X L)}{I_0 \cdot \exp(-\tau X L) + k'_s} \cdot X \right] \quad (9)$$

$$\frac{dS}{dt} + [\nabla \cdot (\mathbf{u}S)] = \nabla \cdot (D_{e,S} \nabla S) + \left[-Y_{XS} \cdot A' \cdot \exp\left(-\frac{E_a}{RT}\right) \cdot \frac{I_0 \cdot \exp(-\tau X L)}{I_0 \cdot \exp(-\tau X L) + k'_s} \cdot X - m \cdot X \right] \quad (10)$$

Table 2

Algorithm pseudocode for coupling light intensity and light attenuation to scalar transport.

Algorithm: Coupling light intensity and light attenuation to scalar transport	
1:	$u_{peak} \leftarrow$ rising plume's maximum velocity
2:	$t_{CFD} \leftarrow$ CFD simulation time
3:	$Y_T \leftarrow$ vertical column's height
4:	$Y \leftarrow$ rising plume's vertical distance
5:	$Y_C \leftarrow$ vertical coordinates of each mesh cell centroid
6:	$I(l) \leftarrow$ light intensity and light attenuation
7:	$C_i \leftarrow$ compartmentalisation of domain into 50. $\sum_{i=1}^{n=50} \left[(i-1) \cdot \frac{Y_T}{50} \ll C_i \ll \left(i \cdot \frac{Y_T}{50} \right) \right]$
8:	for all cells in discretise domain do
9:	for all compartments $C_{i=1 \text{ to } 50}$ in domain do
10:	$Y \leftarrow u_{peak} \cdot t_{CFD}$
11:	if $Y_C > Y$ then
12:	$I(l) = 0.0$
13:	else
14:	$I(l) = I_0 \cdot \exp(-\tau X L)$ see (Anye Cho et al., 2021)

$$\frac{dH_2}{dt} + [\nabla \cdot (\mathbf{u}H_2)] = \nabla \cdot (D_{e,H_2} \nabla H_2) + \left[\phi' \cdot A'_{H_2} \cdot \exp\left(\frac{E_{a,H_2}}{RT}\right) \cdot \frac{I_0 \cdot \exp(-\tau \cdot X \cdot I)}{I_0 \cdot \exp(-\tau \cdot X \cdot I) + k'_{s,H_2}} \cdot X \right] \quad (11)$$

In Eq. (9) to (11), the light saturation coefficients (i.e., k'_s and k'_{s,H_2}) and biohydrogen enhancement factor (ϕ') are intrinsic to the VT-PBR meanwhile the estimated k_s , k_{s,H_2} and ϕ in our previous work (Anye Cho et al., 2021) were based on a smaller PBR (i.e., Schott bottle PBR). Therefore, the conversion from the referenced Schott bottle PBR into the VT-PBR's CFD model warranted the formulation of Eq. (12) and Eq. (13) to embed the consequential effects of light/dark cycles onto the VT-PBR's intrinsic light saturation coefficients. Notice that the parameters on the right-hand side of both Eq. (12) and Eq. (13) are known since the *effective light coefficient* of the referenced Schott bottle PBR, $\eta_{SB} = 1.0$. However, parameters on the left-hand side are unknown but for any given one (e.g., k'_s or η_{VT}) whereby η_{VT} is the VT-PBR's intrinsic *effective light coefficient*, the other was straight forwardly identifiable as with Eq. (14). This implies that both k'_s and k'_{s,H_2} have to be firstly estimated for η_{VT} and η_{VT,H_2} to be identified. However, it is well known in the literatures (Anye Cho et al., 2021; Zhang et al., 2015) that these two intrinsic parameters have similar order of magnitudes (i.e., k'_s , k'_{s,H_2}), therefore only k'_s was to be estimated herein together with ϕ' for the VT-PBR's intrinsic biohydrogen partial pressure influences. This was achieved by minimising the sum of squares errors between the data (i.e., Eq. (4) to Eq. (6)) and the CFD coupled hydrodynamic-biokinetic model (i.e., Eq. (9) to Eq. (11)) resulting to k'_s , ϕ' and η_{VT} as 7.0 ($W\ m^{-2}$), 5.65 (dimensionless), 0.063 (dimensionless) respectively.

$$\left(\frac{k'_s}{\eta_{VT}}\right)_{VT-PBR} = \left(\frac{k_s}{\eta_{SB}}\right)_{SB-PBR} \cdot \frac{\eta_{SB}}{\eta} \quad (12)$$

$$\left(\frac{k'_{s,H_2}}{\eta_{VT,H_2}}\right)_{VT-PBR} = \left(\frac{k_{s,H_2}}{\eta_{SB}}\right)_{SB-PBR} \cdot \frac{\eta_{SB}}{\eta} \quad (13)$$

$$\eta_{VT} = \frac{k'_s}{\frac{499.99}{1.0} \cdot \frac{1.0}{4.51}} = \frac{k'_s}{110.74} \quad (14)$$

Whereby the subscript SB – PBR represents the referenced Schott bottle PBR and η_{VT,H_2} is the *effective light coefficient* for biohydrogen production.

To model the external recirculation of X, S and H_2 : (i) mean residence time was assumed equivalent to one pass through the vertical column and was determined from Eq. (A1) to Eq. (A3) in the Appendix via the pulse input technique as discussed in the work of Zhang et al., (Zhang et al., 2007), (ii) within each pass, the VT-PBR's inlet section was fixed to the value of the outlet's volume (i.e., area $\times$ 1 unit for a 2D CFD) average concentration from the previous pass, (iii) after each pass, the volume average concentration at the VT-PBR's outlet section was recalculated, and this result was patched into the VT-PBR's inlet section via the Dirichlet boundary condition and maintained for the duration of that pass as presented in Table 1. These three steps (i.e., (i), (ii), and (iii)) guarantees the complete recirculation of the bioreaction species from the VT-PBR's outlet back into the inlet section during the time course of the CFD simulation. As well, it renders the pass time accelerable for the growth kinetics thereby decoupling it from the fluid dynamic's simulation time since it cannot be accelerated. The entire coupling strategy was summarised in Fig. 2.

2.2.4. Solver settings and solution strategy

A time step size of 0.005 s and simulation time of 75 s was used to firstly solve the CFD hydrodynamic model to convergence (i.e., residuals drop below 10^{-6} and negligible changes in velocity from one iteration to other for all the monitoring points). Secondly, 6 passes (i.e., $6 \times t_m$ s) were chosen to approximate the entire bioprocess and the same time step size of 0.005 s was considered for the mass (scalar) transport model. The computational cost-saving approach proposed by Anye Cho et al., (Anye Cho et al., 2021) was implemented by scaling three model parameters: A' , m and A'_{H_2} by a factor of $\frac{192\ \text{hrs}}{6 \times t_m\ \text{s}}$, accelerating biomass growth, substrate consumption and biohydrogen production, respectively. As per the parallel computing, the CFD simulations were parallelised over 16 processors. Also, 8, 10 and 12 processors were tested to

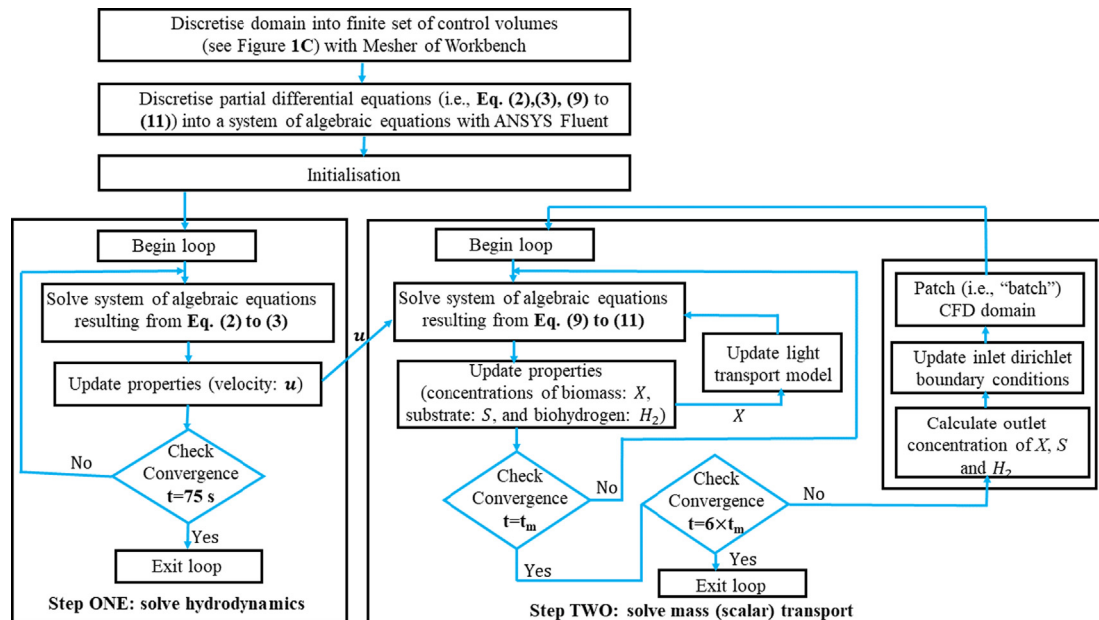


Fig. 2. Schematic representation for the CFD coupling of hydrodynamics to light and biokinetics transport within an externally pumped-recirculated photobioreactor.

analyse for the cost savings due to increasing number of processor cores. Each of the processor cores had 4 GB and the implementation was carried out using the High-Performance Computing (HPC) cluster at The University of Manchester, United Kingdom.

2.2.5. CFD uncertainty propagation

The variability in the coupled hydrodynamic-biokinetic model predictions stems from the uncertainty associated with the estimated model parameters. While the impact of parameter uncertainty on a pure biokinetic model has been well quantified in literature (Anye Cho et al., 2021; Sadino-Riquelme et al., 2020; Anane et al., 2019; Sin et al., 2010) using Monte Carlo techniques, the impact of hydrodynamic coupling on the overall propagated uncertainty remains unknown. Therefore, to investigate this, a 5% input uncertainty was assigned to the transport phenomena related parameters as reported in our previous work (Anye Cho et al., 2021). This corresponds to (i) $\frac{k_s}{\eta}$, $\frac{k_{s,H_2}}{\eta}$, τ and ϕ for the pure biokinetic model, and (ii) k'_s , k'_{s,H_2} , τ and ϕ' for the CFD coupled hydrodynamic-biokinetic model. This choice was motivated by the suitability of these parameters for online model recalibration during process optimisation. While the parameters collectively impact all three state variables, the state variables are also robust to mild perturbations of these parameters, as reported by Anye Cho et al., (Anye Cho et al., 2021). Latin Hypercube Sampling (LHS) drew 100 probabilistic samples from the parameter input spaces whereby each sample was used to simulate one possible

process simulation using either the pure biokinetic model (i.e., Eq (4) to (6)) or coupled hydrodynamic-biokinetic model (i.e., Eqs. (9)–(11)). The former was simple to execute in Python version 3.7 using the SMT 1.0.0, Numpy and SciPy libraries, while the latter was more complex, requiring ANSYS Fluent version 19.2. To grasp the implementation of hydrodynamic-biokinetic modelling in ANSYS Fluent, the readers are encouraged to revisit the works of Anye Cho et al., (Anye Cho et al., 2021).

3. Results and discussion

3.1. Grid sensitivity analysis and CFD fluid flow validations

To compare and validate the CFD model's prediction against experimental results, the simulation was run with four different mesh resolutions (i.e., 50336, 104584, 205682, and 526257 grid elements), each composed of quadrilateral-based elements as shown in Fig. 1 C). As seen in Fig. 3 B), the CFD models predicted the local experimental velocity of all 10 measuring points to within a small percentage relative error (i.e., RE (%)) of less than 10% which is considered as satisfactory in other literature for PBR validation studies (Huang et al., 2016; Xu et al., 2014; Cho and Pott, 2019). This confirms the reliability of the 2D CFD model for approximating the hydrodynamics of the 3D VT-PBR. Although increasing the number of grid elements from 50,336 to 526,257 did not improve the CFD's numerical accuracy, the computational time cost was observed to increase from 0.57 hrs to 0.68 hrs and

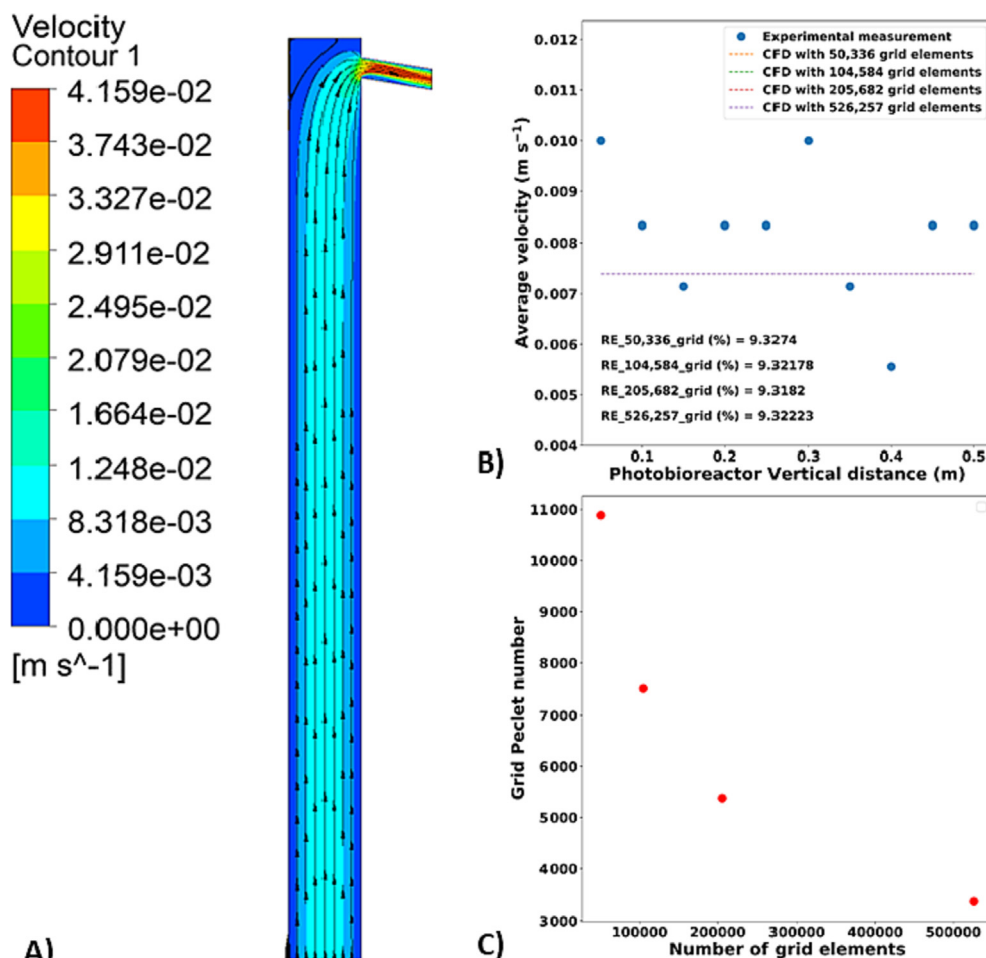


Fig. 3. Results of A) CFD velocity contours with post-processed arrows indicating flow pattern and directions, B) CFD validation at monitoring points and grid sensitivity analysis with their indicated %RE, overlay of the CFD results due to similar %RE, and C) influence of the number of CFD grid elements on the grid Peclet number.

0.91 hrs, respectively. This result demonstrates that using the smallest number of grid elements (i.e., 50,336 grid elements) were sufficient for accurate CFD hydrodynamic prediction while minimising the computational time cost by 37.4 % compared with the highest 526,257 grid element option.

However, C) of Fig. 3 shows the grid Peclet number (P_e) to decreases with an increasing number of grid elements from 50,336 to 526,257, resulting in the highest and lowest P_e values of 10,884.2 and 3,359.5 respectively. This agrees with other literature findings (Liu, 2011; Bayareh, 2021) but high P_e values have serious consequences for numerical stability by inducing numerical (i.e., false) diffusion, which often masks the fluid's molecular diffusion, thereby inducing truncation errors into the advection–diffusion transport equations (Liu, 2011; Michalcová and Kotrasová, 2020; Bayareh, 2021). Therefore, given the stiff nature of the ordinary differential equations (ODEs) of the source growth terms embedded into the mass (scalar) transport equations, the CFD model with the highest number of grid elements (i.e., 526,257) and lowest P_e was selected for all subsequent analysis within this work.

3.2. CFD hydrodynamic insights

As the CFD model has been validated against experimental data, its full potential can be unleashed to explore the VT-PBR's hydrodynamic modes of operations, in particular: (i) the flow profile, and (ii) areas of poor mixing or hydrodynamic dead zones. Beginning with the flow profile, the CFD hydrodynamic model predicted an average velocity of $7.7 \times 10^{-3} \text{ m s}^{-1}$ with a corresponding light fraction time of 0.7 (T_L see Eq. (7) and Eq. (8)). As well, a parabolic flow profile exhibited by the arrowheads within the path lines across A) of Fig. 3 are indicative of a laminar flow regime, thus validating the choice of the laminar flow regime during the CFD simulation solver setup as indeed correct. Now looking at hydrodynamic mixing within the VT-PBR, it can be seen that mixing was axially dominated, reflected by the vertically pointing arrowheads in Fig. 3 A), which maintained a parallel trajectory from the inlet to the outlet. However, as visualised by the deep blue regions of the contour plot, hydrodynamic dead zones with lower velocities occurred near the walls and top of the vertical column. Low fluid velocities near the walls were expected, given the non-slip boundary condition at the walls of the vertical column in addition to the parabolic profile of the laminar flow which mainly peaks at the centre and then decreases towards the walls. Therefore, these results reveal the need to improve radial mixing within the VT-PBR to alleviate the stagnant near-wall regions to minimise prolonged exposure periods of either light intensity (i.e., photoinhibition) or darkness (i.e., photolimitation). Some have reported that static mixers can improve radial mixing; however, while many designs have been proposed for flat-plate PBRs (Wang et al., 2021; Wang and lin, Tao Y, Mao X zhong, 2014; Huang et al., 2015; Yu et al., 2009; Münkkel et al., 2013), very few designs have been proposed for tubular PBRs (Salguero-Rodríguez et al., 2021; Perner-Nochta and Posten, 2007; Qin et al., 2018).

3.3. Parameter estimation and CFD simulation results

To visualise the performance of the current photobioreactive transport modelling strategy, the CFD predicted light distribution and biomass concentration contours at four different simulation times within the 1st pass were plotted in Fig. 4. It is seen that at the start (Fig. 4 A)), the light distribution is zero through the domain and as the simulation progresses, the light distribution only becomes non-zero (Fig. 4 B), C) and D)) for regions harboring

the rising plume as indicated with the dash lines. Similar observations were made for the other coupled state variables (i.e., substrate consumption and biohydrogen production) and in the subsequent passes (i.e., 2nd to 6th) with the results not shown herein, thus validating the proposed algorithm. However, beneath the dash lines in Fig. 4 B) to C) shows the algorithm to be missing out the curvature profile of the laminar plume, thereby activating the light transport model for a few more mesh cell elements which are unoccupied by the rising scalar, presenting an area of the algorithm's improvements.

Fig. 5 A), C) and E) shows the model fitting and simulation results, where the percentage relative errors (i.e., RE (%)) for the biomass, substrate and biohydrogen models were 21.0 %, 7.1 % and 24.4 %, respectively. This high RE (%) were mostly observed for the intermediary passes (i.e., 2nd to 5th passes) and the good comparison at the 6th pass suggests decreasing RE (%) with increasing number of simulated passes. However, the improved performance achieved by increasing the number of passes comes at the cost of increased computational CFD simulation time. Therefore, a good compromise between simulation accuracy and computational cost must be struck. The 6 passes herein took 22.28 hrs, 27.22 hrs, 30.3 hrs, and 36.87 hrs of wall clock time to simulate 508.14 s of CFD simulation time when parallelised over 16, 12, 10 and 8 processor cores respectively. As expected, the cost savings scales (i.e., simulation runs faster) with increasing number of processor cores. However, the speed-up was not linear as doubling the processor cores from 8 to 16 did not reduce the wall clock time by half. This suggests the speed-up to tail off beyond 16 processor cores, and greater than 16 cores were therefore not investigated.

Of the 22.28 hrs wall clock time, it only took 0.91 hrs to simulate the hydrodynamics in Step ONE; the remaining 21.38 hrs were spent simulating the three state variables in Step TWO. Hence the model in this work is very efficient compared to others in the literature that took several days to converge even for a single state variable (Nauha and Alopaeus, 2015; Amini et al., 2018) using larger time step sizes of 3600 s (Amini et al., 2018) and 10 s (Gao et al., 2017). Therefore, while increasing the number of passes would incur an additional computational cost, the resulting clock time would remain competitive with published works, given the high accuracy achieved already with only 6 passes.

3.4. Uncertainty propagation results

The uncertainty bands in Fig. 5 B), D) and F) shows the degree of variability in the predictions from the biokinetic model parameters into the CFD coupled hydrodynamic-biokinetic model as well as the pure biokinetic model in Appendix B. For all state variables, the propagated uncertainty increased with time, demonstrating that the models were responsive to changes in the parameter input space while not being too sensitive, in agreement Anye Cho et al., (Anye Cho et al., 2021). Comparing the two models, the uncertainty of the pure biokinetic model approaches that of the coupled hydrodynamic-biokinetic model towards the end of the last (i.e., 6th) pass. This similarity suggests that the CFD hydrodynamic to biokinetic coupling was robust to false diffusion related errors. Also, the comparable percentage deviation between the two mean Monte Carlo predictions in Fig. 5 B), D) and F) to those of Appendix B (i.e., without Monte Carlo) suggest that both models will perform the same under mild level of uncertainty provided that the solution quality in Section 3.3 was improved (i.e., minimise the high %RE in intermediary passes). Overall, this implies that the mathematically derived hydrodynamic and transport phenomena related parameters in the literature (Anye Cho et al., 2021; Anye Cho et al., 2021), and the formulated Eq. (12) to Eq. (14) for estimating the VT-PBR's intrinsic parameters were theoretically sound, meaning

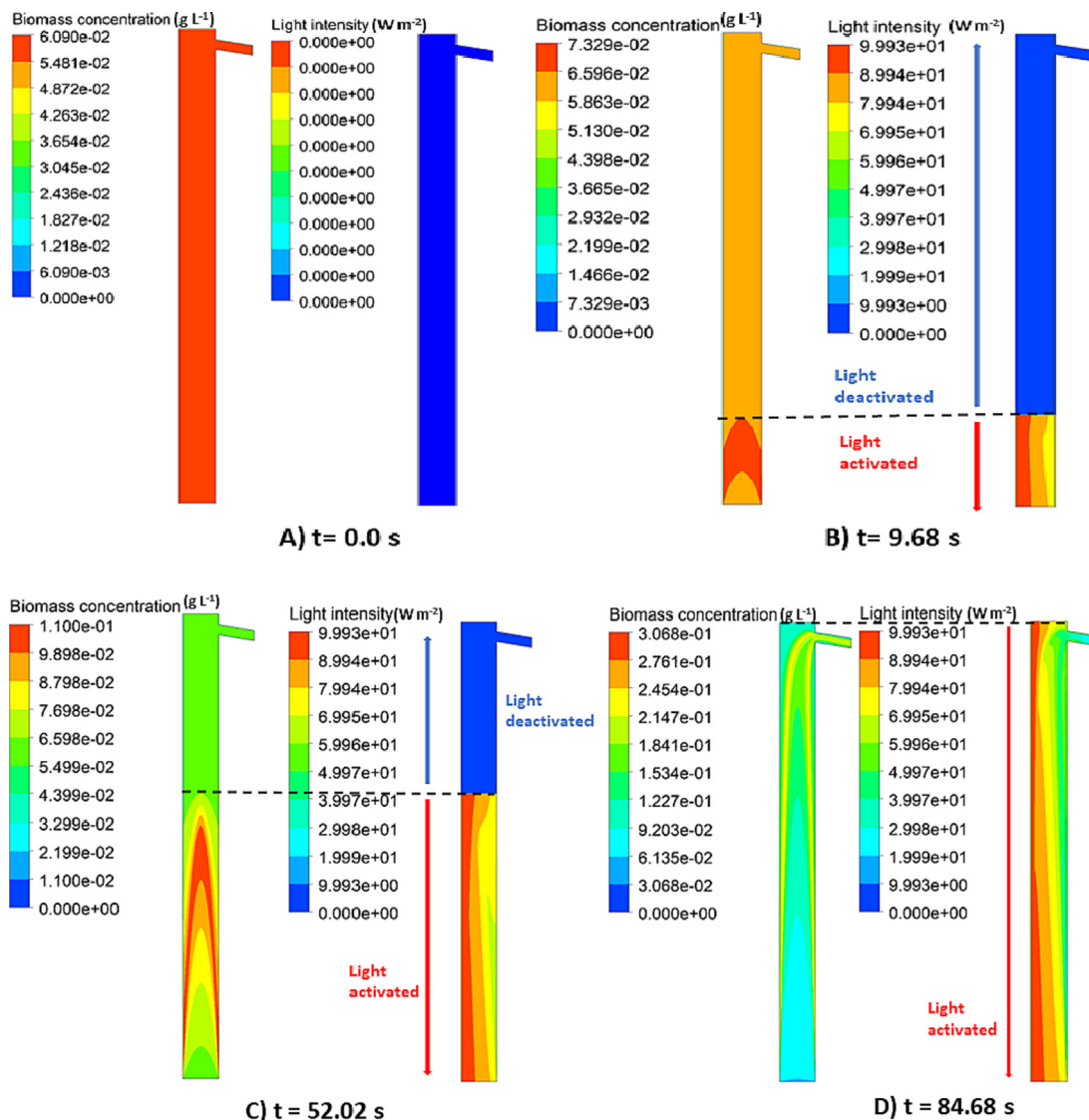


Fig. 4. CFD prediction results of biomass and light intensity distribution at various time instances during the 1st pass. A) at the start, B) 9.68 s, C) 52.02 s, and D) end of pass. No photobioreactions outside of rising plume is indicated with blue arrows and the source term was always zero in the outlet pipe section.

they can be perturbed directly to investigate the hydrodynamic influence on the overall PBR's productivity.

Also, the influence of the CFD coupled hydrodynamic-biokinetic model's uncertainty propagation on the local state variables (e.g., distribution of biomass concentration) and growth limiting factors (i.e., local light intensity) were further analysed with the lower and upper uncertainty bands at the end of the final pass. This was since the literature studies (Zhang et al., 2015; del Rio-Chanona et al., 2019) considered optimisation of the biomass concentration at the final cultivation period. For this, the column was vertically split up into 5 equal sections (0.01 m each) along the PBR's light path length (a scalable variable) and the volume averages of the local biomass concentration and local light intensities reported respectively in Fig. 6 A) and B). The results showed (i) slight inverse correlation between local biomass concentration and local light intensity, (ii) strong coupling between local biomass concentration and light intensity as they both decrease with increasing PBR's diameter, (iii) local light intensity was critically low (i.e., $<5 \text{ W m}^{-2}$) beyond 0.02 m. Meanwhile the implications of (ii) and (iii) for the design and upscaling of PBRs have already been dis-

cussed in the literature (Socher et al., 2016; Richmond, 1996), (i) is revealed herein as an interacting effect which could increase to significant levels with increasing uncertainty bands. Hence, the PBR's optimisation under uncertainty might guarantee a good compromise between the uncertainties's upper and lower bands while not violating this inverse correlation for the optimal solution.

4. Conclusion

This paper presents a cost-efficient CFD based framework exploiting parallelised computing environment, and cost-saving accelerated growth kinetics, for the modelling of light and bioreaction species transport in a vertical tubular photobioreactor under uncertainty. The simulated (i) velocities and (ii) bioreaction species (i.e., biomass growth, substrate consumption and biohydrogen production) agreed well with the experimental data to within 10% and 24% relative errors, respectively. However, the absence of radially dominated flow patterns in the simulated velocities suggests the need for the PBR's structural optimisation to eliminate stagnant

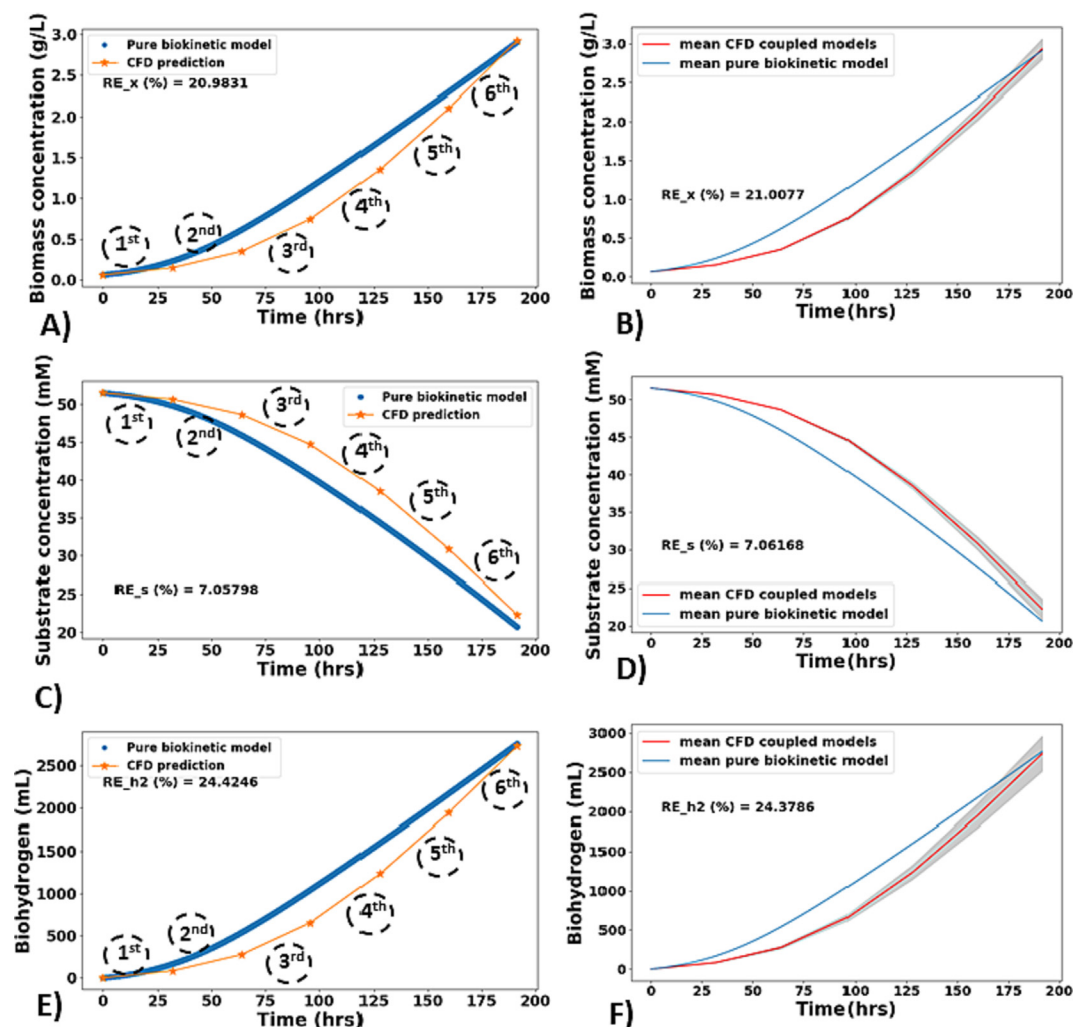


Fig. 5. CFD end of pass prediction for the column's volume average results for A) biomass growth, C) substrate consumption, and E) biohydrogen production compared to the respective pure biokinetic simulation as the data. B), C) and F) shows CFD uncertainty propagation results for the state variables. The numbers enclosed within broken circles denote the pass number and the asterisk demarcates the end of that pass. The percentage relative errors (%RE) between the CFD and actual biokinetic predictions are indicated in each plot.

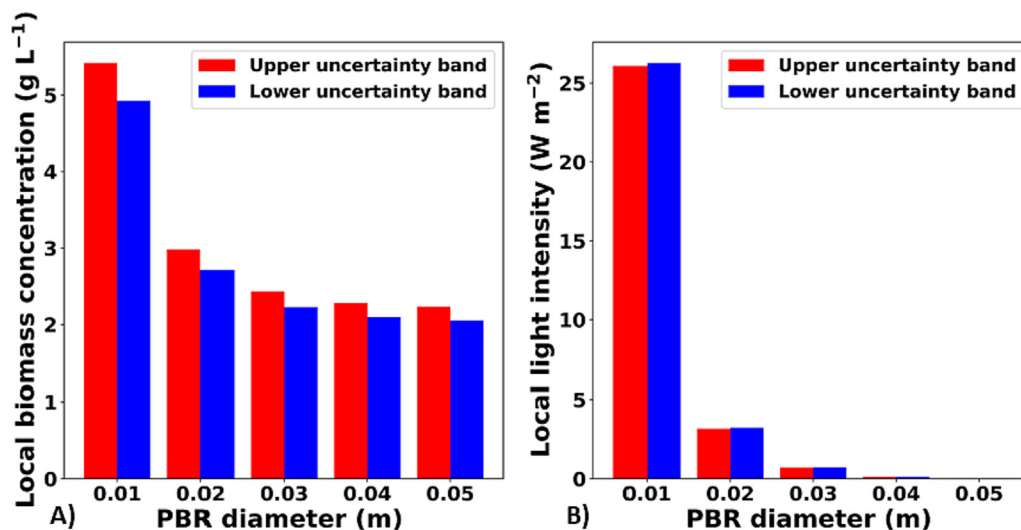


Fig. 6. Effects of uncertainty propagation in the CFD coupled hydrodynamic-biokinetic models at the end of the final pass. The x-axis of A) and B) corresponds to volume averages for 0.01 m intervals through the vertical height of the vertical column for the lower and upper uncertainty bands respectively.

near-wall regions. Regarding the increasing accuracy for the simulated bioreaction species with increased number of simulated passes, it has been found that there is a trade-off between the simulation accuracy and computational cost. Therefore, optimisation of the PBR's flow and biohydrogen productivity under uncertainty are now computationally feasible.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The author Mr. B. Anye Cho is fully funded by the Commonwealth Scholarship Commission, United Kingdom.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ces.2023.118525>.

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